BETOBULLET

14 HP · EURORACK FORMAT
FOR VCV RACK 2



FELONY

ORNAMENT & CRIME FOR VCV RACK

USER MANUAL · v2.0.0 · October 2026

CONTENTS

1. About Felony
2. Panel overview
3. Controls
4. The apps
5. Saving and VCV specifics
6. Specifications

Felony runs the real Ornament & Crime firmware on a virtual O&C: quantizers, envelope generators, LFOs, sequencers, chaos and bytebeats, exactly as on the hardware, screen included.

1. ABOUT FELONY

Ornament & Crime is an open source module (mxmxmx/O_C, MIT licence) whose firmware hosts a collection of "apps". Felony compiles that firmware unchanged and runs it on a virtual O&C: the 128×64 OLED, the DAC8565 outputs, the ADC inputs, the trigger inputs, the encoders, buttons and EEPROM are emulated. What you see on the screen and hear at the outputs is what the hardware would do.

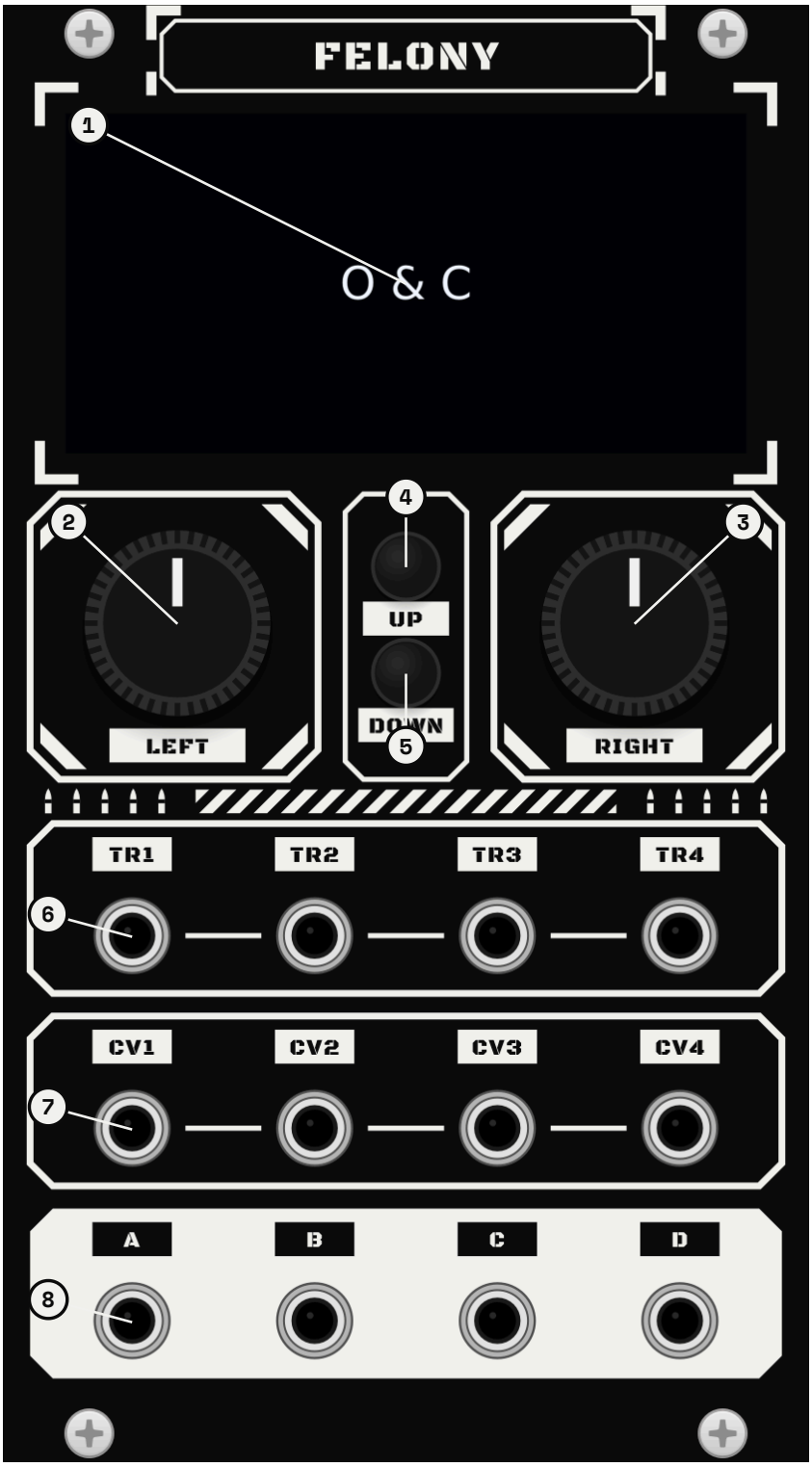
Because the apps are the original ones, the official O&C user manual (`ornament-and-cri.me/user-manual-v1_3`) applies to every app in full. This document covers the module, its controls and what is specific to VCV Rack.

INSTALLATION

Copy `Felony-2.0.0-win-x64.vcvplugin` to `Documents/Rack2/plugins` and restart VCV Rack. The module is listed under the brand **betobullet**.

ONE PER RACK. The firmware keeps its state in global variables, so only one Felony can run in a VCV Rack session. A second Felony shows a notice on its screen and stays inactive.

2. PANEL OVERVIEW



1. **Screen**
   The 128×64 OLED of the O&C, drawn from the firmware's own display memory.
2. **LEFT encoder**
   Turn, click and long press. Usually selects a channel or a parameter.
3. **RIGHT encoder**
   Turn, click and long press. Usually edits values; long press opens the app list.
4. **UP button**
   App-specific; long press toggles the screensaver.
5. **DOWN button**
   App-specific.
6. **TR1–TR4**
   Gate / trigger inputs (threshold about 1 V).
7. **CV1–CV4**
   CV inputs, −3.3 V to +6.6 V.
8. **A–D**
   Outputs, −3 V to +6 V, exact 1 V/oct calibration.

3. CONTROLS

ENCODERS WITH A MOUSE

ACTION	MOUSE
Turn	drag up/down or scroll the wheel
Click	click and release
Long press	click and hold still for one second

The encoders accelerate exactly like the hardware: turn slowly for single steps.

SELECTING AN APP

1. **Hold the RIGHT encoder** for about one second: the app list appears.
2. **Turn the RIGHT encoder** to the app.
3. **Click** to start it, or **hold** to start it and save it as the power-on app.

4. THE APPS

APP	WHAT IT DOES
CopierMaschine	Analogue shift register: sample & hold with a quantized four-step delay line.
Harrington 1200	Neo-Riemannian transformations of triads, driven by triggers.
Automatonnetz	Sequencer walking the Tonnetz grid of chords.
Quantermain	Quad quantizer with scales, Turing machines and integer sequences.
Meta-Q	Dual quantizer that sequences its own scales.
Quadraturia	Quadrature wavetable LFO (from Mutable Instruments Frames).
Low-rents	Lorenz and Rössler strange attractors.
Piqued	Four envelope generators with a graphic envelope monitor.
Sequins	Dual step sequencer.
Dialectic Ping Pong	Bouncing-ball envelopes.
Viznutcracker sweet	Bytebeat generator.
Acid Curds	Chord progression sequencer and quantizer.
References	Reference voltages and a frequency counter (tuner) on TR4.

Full documentation of every app: the official O&C user manual, version 1.3.

5. SAVING AND VCV SPECIFICS

SAVING

On the hardware, settings survive power-off only after a long press in the app list, which writes them to the EEPROM. Felony does that automatically whenever the patch is saved: the current app and the settings of **all** apps are written to the virtual EEPROM, and the EEPROM is stored in the patch. Reopening the project brings Felony back in the same app with the same settings.

As on the hardware, momentary state is not saved: a sequence position or an envelope in progress starts fresh.

CONTEXT MENU

ITEM	FUNCTION
Screen colour	White, Blue, Yellow or Green OLED
Theme	Black or White panel
Power cycle	Restarts the firmware, like switching the module off and on

VOLTAGES

Inputs and outputs use the ranges of the hardware: CV inputs read −3.3 V to +6.6 V, outputs reach −3 V to +6 V. Calibration is ideal: 1 V/octave is exact without running the calibration routine.

6. SPECIFICATIONS

ITEM	VALUE
Width	14 HP
Firmware	Ornament & Crime (mxmxmx/O_C), unchanged apps
Display	128 × 64 OLED, emulated SH1106
Inputs	4 × trigger/gate, 4 × CV (−3.3…+6.6 V)
Outputs	4 × CV (−3…+6 V), 16-bit
Core rate	16.666 kHz, as hardware
Storage	EEPROM saved in the patch
Limit	one instance per Rack session

CREDITS

Ornament & Crime firmware by Max Stadler, Patrick Dowling, Tim Churches and contributors (MIT). Virtual hardware, panel and VCV layer: betobullet. Fonts: Black Ops One, Space Mono, Space Grotesk (SIL OFL).